

Symbolic Grounding Reveals Representational Bottlenecks in Abstract Visual Reasoning

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Abstract

Vision–language models (VLMs) often fail on abstract visual reasoning benchmarks such as Bongard problems, raising the question of whether the main bottleneck lies in reasoning or representation. We study this on Bongard-LOGO, a synthetic benchmark of abstract concept learning with ground-truth generative programs, by comparing end-to-end VLMs on raw images with large language models (LLMs) given symbolic inputs derived from those images. Using symbolic inputs as a diagnostic probe rather than a practical multimodal architecture, our *Componential–Grammatical (C–G)* paradigm reformulates Bongard-LOGO as a symbolic reasoning task based on LOGO-style action programs or structured descriptions. LLMs achieve large and consistent gains, reaching mid–90s accuracy on Free-form problems, while a strong visual baseline remains near chance under matched task definitions. Ablations on input format, explicit concept prompts, and minimal visual grounding show that these factors matter much less than the shift from pixels to symbolic structure. These results identify representation as a key bottleneck in abstract visual reasoning and show how symbolic input can serve as a controlled diagnostic upper bound.

1 Introduction

Vision–language models (VLMs) have achieved strong results on captioning, retrieval, and recognition from large-scale paired data (Radford et al., 2021; Jia et al., 2021; Ramachandran et al., 2025), yet they often fail on tasks that require abstract reasoning, compositional generalization, or spatial inference (Yuksekgonul et al., 2022; Parascandolo et al., 2025; Li et al., 2025). Analyses of CLIP-style encoders suggest that end-to-end architectures may overemphasize salient objects or early caption tokens while underrepresenting relational and structural information (Fine and Abend, 2025; Kang and

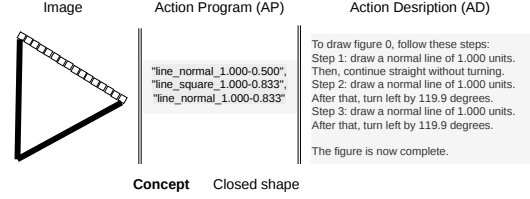


Figure 1: Example Bongard-LOGO instance under the Componential–Grammatical interface. From left to right we show the rendered shape, its action-program (AP) representation, and its action-description (AD) rendering. In the concept-conditioned variant, the prompt additionally provides the high-level concept label, allowing us to compare visual, procedural, and natural-language interfaces for the same underlying example.

Choi, 2023; Campbell et al., 2024). This raises a basic question: is the main bottleneck in current multimodal models their *reasoning* mechanisms, or the *representations* that feed those mechanisms?

We address this question in a controlled synthetic setting: **Bongard-LOGO** (Nie et al., 2020), a benchmark of concept-learning tasks inspired by classic Bongard problems (Bongard, 1970; Foundalis, 2006). Each problem presents positive and negative example images and requires inferring a latent rule that separates them. Figure 1 previews the three interfaces used in our experiments: the rendered image itself, its procedural action program, and its natural-language action description. Unlike conventional benchmarks, Bongard-LOGO emphasizes geometric, relational and topological properties—symmetry, convexity, connectivity—rather than object semantics, and exposes the ground-truth procedural LOGO program that generates each image.

Classic work on Bongard problems argues that solving them requires a *visual language* of structured primitives and relations; Bayesian and causal neuro-symbolic systems define such vocabularies and inference procedures, achieving strong performance on hand-designed sets while remaining in-

interpretable (Depeweg et al., 2024; Grinberg et al., 2023). We follow this line but replace bespoke engines with general-purpose language models as the reasoning backend.

We introduce the **Componential–Grammatical (C–G)** paradigm: a modular interface that decouples perception from reasoning by posing Bongard-LOGO as a symbolic reasoning problem. Instead of operating on pixels, models receive the underlying *action programs* that generated the images or structured textual descriptions derived from them, forming a LOGO-style visual language. A language model then induces a rule and classifies query examples from symbolic input alone. This setup lets us ask: to what extent do recent language models exhibit abstract generalization when supplied with explicit symbolic structure, and how do representation format, concept conditioning, and minimal visual grounding modulate their performance?

Empirically, a strong VLM baseline (Gemini-2.5-Flash) performs near chance ($\sim 50\%$) on Bongard-LOGO, whereas the same tasks reframed via symbolic input yield substantially higher accuracies (often 60–80% across models, with Phi-4-Reasoning reaching 96.2% on Free-form problems). The main driver of this gap is representational: compositional, interpretable structure enables generalization, while variations in format, concept prompts, and grounding yield smaller, model-dependent effects. Although symbolic input is not a realistic perceptual modality, we treat C–G as a diagnostic upper bound on what current models can do once grounded in an appropriate visual language, rather than as a complete solution to perception.

Our goal is diagnostic rather than architectural. We do not introduce a new end-to-end multimodal system or learned visual front-end; instead, we vary the *representational interface* between perception and reasoning on Bongard-LOGO. By comparing the same family of language models under procedural programs, natural-language descriptions, concept prompts, and structure-breaking perturbations, we isolate how input format and inductive bias shape abstract generalization independently of vision encoder choice.

Concretely, our contributions are threefold. *First*, we present the Componential–Grammatical (C–G) interface for Bongard-LOGO and show that representational format alone can induce 25–30 point gains over a strong visual baseline across 12 models. *Second*, we systematically probe this interface

via concept conditioning, minimal-context prompting, grounded variants, and token randomization controls, revealing when and how symbolic structure—rather than surface statistics—drives success. *Third*, we provide a cross-model analysis of capacity and brittleness under these interfaces, positioning C–G as a diagnostic tool for studying representational bottlenecks in multimodal reasoning.

While prior work has shown that specialized symbolic solvers perform well on Bongard problems, it is not obvious that general-purpose pre-trained LLMs can induce abstraction rules directly from procedural symbolic traces without domain-specific search or operators. Our experiments show that once grounded in an explicit visual language, modern LLMs can function as effective abstract inference engines.

2 Related Work

Vision–Language Models and Grounding.

Modern VLMs such as CLIP and ALIGN learn joint image–text representations and excel at recognition and retrieval (Radford et al., 2021; Jia et al., 2021; Ramachandran et al., 2025), but benchmarks targeting compositional generalization, spatial reasoning, and relational understanding reveal systematic failures (Yuksekgonul et al., 2022; Ma et al., 2023; Hu et al., 2023; Hsieh et al., 2023). These models often prioritize coarse object or token statistics, with limited sensitivity to attribute binding, part–whole structure, or word order, and analyses trace this to architectural and training biases such as overreliance on early caption tokens or spatial salience (Fine and Abend, 2025; Kang and Choi, 2023; Parascandolo et al., 2025; Campbell et al., 2024). Recent work introduces more explicit intermediate structure, via multi-granular alignment between sentence parts and image regions (Le et al., 2025), generative-flow-style chain-of-thought (CoT) reasoning over images (Kang et al., 2025), or program-like plans over symbolic state (Xu et al., 2025). In contrast, we change only the *input format*—from pixels to LOGO-style symbolic programs—and ask how this affects abstract generalization in language models.

Bongard Problems and Visual Abstraction.

Bongard problems are classic visual reasoning tasks where a rule must be induced to separate positive from negative examples (Bongard, 1970; Hofstadter, 1979; Foundalis, 2006). They empha-

size geometric and topological concepts such as symmetry, containment, and convexity, and have informed work on analogy and visual concept learning (Gentner and Holyoak, 1997; Gentner and Hoyos, 2017). Bongard-LOGO (Nie et al., 2020) extends this paradigm into a synthetic benchmark of 12,000 procedurally generated problems from ground-truth LOGO programs. Despite their visual simplicity, these tasks are difficult for current models: even advanced systems like GPT-4o and Gemini attain only 50–60% accuracy on abstract Bongard-style sets, far below human performance (Wüst et al., 2024). Several systems treat Bongard problems as inference in an explicit visual language: Depeweg et al. (2024) define a symbolic vocabulary and use Bayesian inference with pragmatic constraints, and Bongard Architecture implements a causal, operator-based neuro-symbolic system that solves nearly all Maksimov–Bongard puzzles while remaining interpretable (Grinberg et al., 2023). Bongard-in-Wonderland shows that frontier foundation models still often fail on simple geometric concepts like spirals (Wüst et al., 2024). Our work follows this tradition in using a structured visual language, but replaces bespoke inference engines with general-purpose LLMs and leverages Bongard-LOGO’s generative programs as the underlying representation.

Symbolic Interfaces and Neuro-Symbolic Models. Our approach fits within a broader effort to modularize perception and reasoning through structured interfaces. Componential Analysis (CA) (Vaishnav and Tammet, 2025) uses a VLM to generate natural-language scene descriptions that an LLM then uses for classification or inference; this works well on natural-image Bongard datasets such as Bongard-OpenWorld (Wu et al., 2024), but performs poorly on Bongard-LOGO, where precise relations are hard to verbalize (Peng et al., 2024). Inverse-graphics and shape-grammar approaches similarly emphasize program-like representations of visual structure (Stiny and Gips, 1972; Gips, 1975; Knight and Stiny, 2015; Kulkarni et al., 2015; Yildirim et al., 2020; Kulits et al., 2024). For Bongard-LOGO, probabilistic concept models combined with Sinkhorn distances over symbolic shapes (Song and Yuan, 2024) and ARC-style neuro-symbolic agents that search in task-specific languages (Batorski et al., 2025) illustrate the value of explicit languages as grounding interfaces. Our Componential–Grammatical (C–G) interface in-

stantiates this pattern for LOGO-based Bongard problems with LLMs as the reasoning backend and allows us to evaluate how procedural programs versus descriptive text modulate downstream reasoning performance independently of perceptual noise.

3 Method

Bongard-LOGO as a Diagnostic Benchmark.

We adopt the LOGO benchmark (Nie et al., 2020) as a controlled testbed for studying the interface between perception and reasoning. Each of its 12,000 problems consists of six positive and six negative example images, along with a query image to classify. All images are procedurally generated from known LOGO programs, enabling precise control over visual properties. Following the released benchmark splits, we report results for *Free-form* (FF), *Basic* (BD), and two *Human-designed* evaluation splits: HD-Comb and HD-Novel. FF concepts are defined by stroke sequences, BD concepts by reusable shape categories and their combinations, and the two HD splits test transfer to human-designed shapes under combinatorial and novel-shape regimes. Unless otherwise noted, HD denotes the mean of HD-Comb and HD-Novel.

Symbolic Representation via Procedural Grammars.

Unlike typical image benchmarks, Bongard-LOGO provides access to the full generative trace for each image. Each figure is constructed from a hierarchy of LOGO-like drawing instructions: a BongardImage comprises one or more OneStrokeShape objects, each built from BasicAction primitives (e.g., Line, Arc). We encode these structures as compact grammar strings that preserve the sequence of operations, angle information, and shape identifiers. For instance, a triangle might be represented as a sequence of three equal-length line segments with 120-degree turns between them. These symbolic traces form a procedural shape grammar that is both noise-free and compositional, aligning with inverse-graphics views of visual processing (Stiny and Gips, 1972; Knight and Stiny, 2015; Kulkarni et al., 2015; Kulits et al., 2024).

Componential–Grammatical Paradigm (C–G).

We introduce the **Componential–Grammatical** (C–G) paradigm, a modular pipeline that separates visual structure recovery from downstream reasoning. C–G operates in two stages: